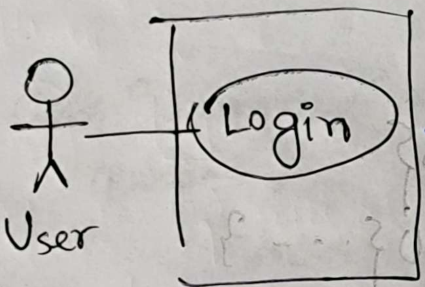
CSE 317
System Analysis & Design
02/02/2026

1. Use case diagrams vary from system to system
   - ↳ The more complexity of project
     - ↳ The more comprehensive use case diagram.

Actor : External Entity
Usecase : Internal Entity
Usecase must be inside boundary
Actor must be outside boundary

5 marks per diagram (use case diagram)



(last topic of this chapter)

Use Case Scenario

→ একটা use caseর documentation
যেখানে,
actorদের ভূমিকা,
use case,
how to make successful use case
backup usecase if it fails

আবার,
usecaseর backup
- ↳ base case (successful scenario)
- ↳ alternative case (alternative scenario)

→ description of use case.

Parts of Use case Scenario

1. Area Header: includes initiators & identifiers
2. Steps performed: steps required to successfully execute the use case
3. A footer:
   1. Preconditions
   2. Assumptions
   3. Questions "Outstanding Issues"
   4. Othe information

Header Area

1. Name & UID
2. Application area
3. Actors & Stakeholders
4. Usecase level ** (later)
5. Description of use case

"Benefits of Use case Diagram" from slide

Use case level

White (clouds) : Enterprise
Kite: Business/Department (HR)
Blue (sea) : User goals
Indigo (Fish): Functional or Sub-functional (OTP generation)
Black (Calm): Most detailed (Technical level)

Chapter III
Project Management

- 'Feasibility Study' ← VVI
- Project Scheduling ☑ {Done in Lab}

An organized sequence of task that satisfies a specific goal is called a project.

→ Definition

Gantt Chart ☑

PERT Diagram {critical}

↳ Program Evaluation & Revenue Techniques

↳ Identifies critical activities

↳ Total project completion timeframe depends on these activities.

~~Sla~~

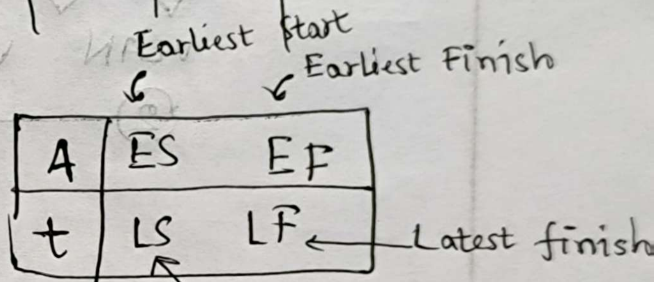
↳ Slack time determination

↳ কোন task কতটুকু delay করা যাবে

PERT Diagram

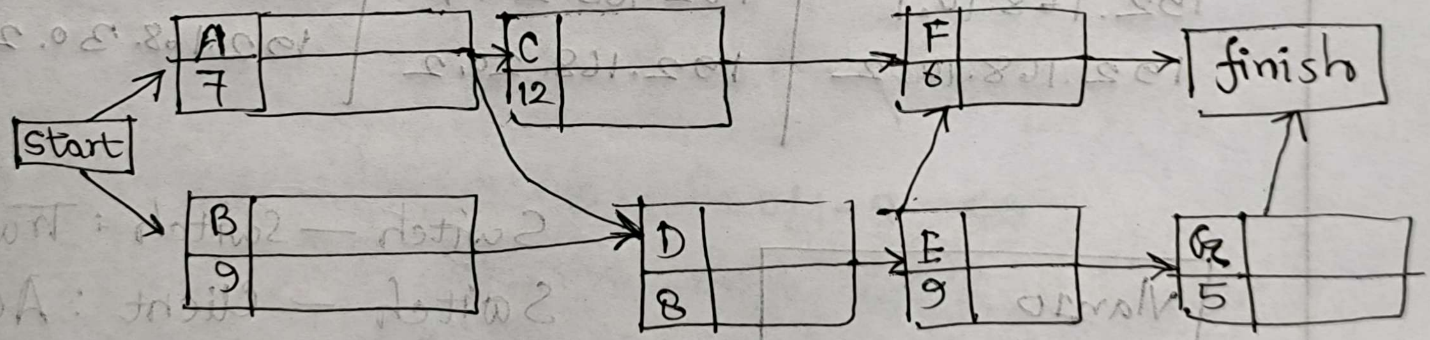
1. Draw the network
2. Determine slack time
3. Critical activities and path

Activity	A	B	C	D	E	F	G
Predeccessor	—	—	A	A,B	D	C,E	E
Expected Time (Weeks)	7	9	12	8	9	6	5



$EF = ES + t$

$LS = LF - t$



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System Analysis and Design

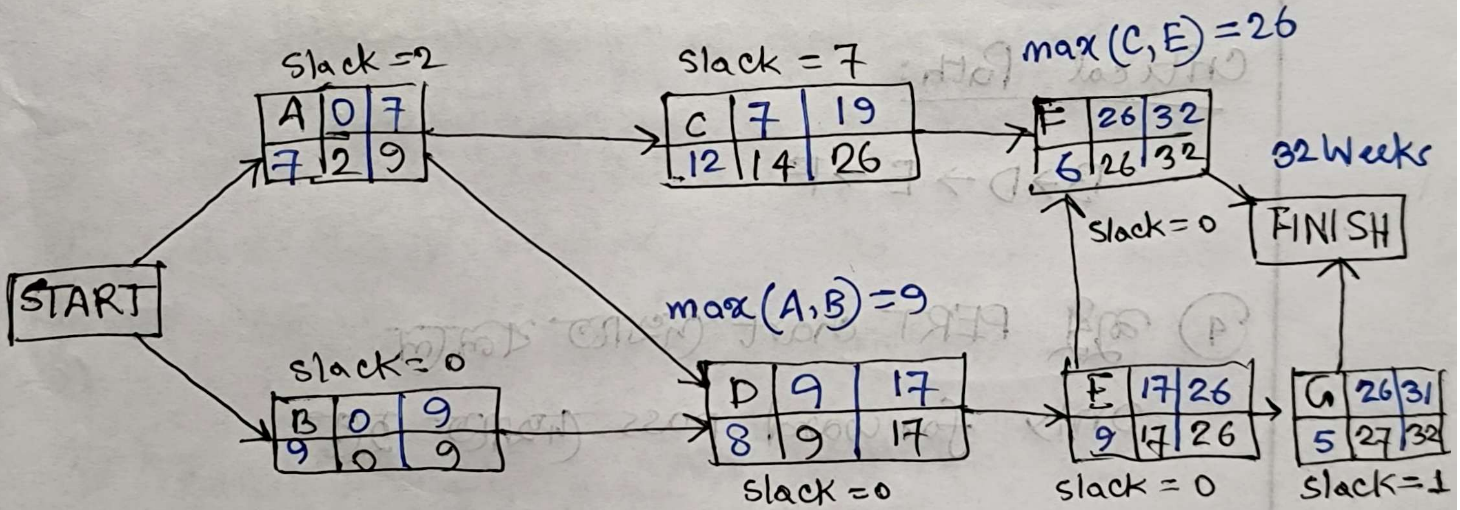
03/02/2026

Chapter III

Project Scheduling (PERT Diagram)

Activity	A	B	C	D	E	F	G
Predecessor	—	—	A	A,B	D	C,E	E
Expected Time (Weeks)	7	9	12	8	9	6	5

PERT Chart Network Diagram



1. Activities with no predecessors can be started parallely.
2. Activities that are not predecessor of any other activity will be connected to the 'finish' node

Next: Slack Time & Critical Path

A	ES	EF
t	LS	LF

Activity → Time

$LS = LF - t$

Forward Pass / Backward Pass

$EF = ES + t$

TPT = total project completion time → (F) took longest to finish

"TPT = 32 Weeks"

③ Je nodeগুলো 'finished' এর ~~sa~~ সাথে connected সেগুলোর LF = TPT.

Slack Time / Slack Value

LF − EF or LS − ES

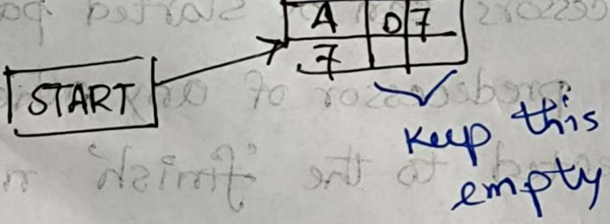
Next: Critical Activities

যাদের "slack time = 0"

Critical Path:

B → D → E → F

④ শুধু PERT chart দেখাতে বললে only forward pass দেখাতে হবে।



Ⓐ What is a project?

A project is a collection of organized task that can be performed either in sequential or parallel manner to achieve a specific goal.

Ⓑ What are the reasons to initiate a project?

① Facing problems in the current infra-structure.
② Opportunities to improve, installing new systems.

Ⓒ How to identify if that's the case??

① "ASK FOR FEEDBACK"
② "Check for performance,
Current Sol^n vs New Sol^n"

③ "Employees going wild"
↳ Job turnover
↳ Job dissatisfaction

Define: Problem

① Problem Statement
↳ Section showing the overall situation of what is going wrong and why attention is needed

② Issues
↳ Specific problems or symptoms

③ Objectives
↳ What the organization want to achieve to solve the issue.

Book: Page 59
'Catherine's Catering' Scenario. ← বাসায় দেখবো
↳ What did Kate do wrong??

Issues ⟶ Weight
{ On the scale of 1 to 10 }
↳ lol, Baymax??

'Next: Feasibility Study'